



E.G.S. Pillay Engineering College

An Autonomous Institution Affiliated to Anna University, Chennai | Approved by AICTE, New Delhi

Accredited by NBA T1 (B.E. – CIVIL, CSE, ECE, EEE, MECH & B.Tech – IT)

Accredited by NAAC with A++ Grade | One among Top 300 Engineering Institutions in India (NIRF-24)

Old Nagore Road, Thethi, Nagore Village, Nagapattinam – 611002, Tamil Nadu, India

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

2025-26 Odd Semester

MODERN PEDAGOGY VERIFICATION

S. No	Faculty Name with Designation	Date of Visit	Class / Section details	Subject name	Day order / Hour	Module details	Pedagogy used
1	Dr.M.Nuthal Srinivasan	24.10.2025	II AIDS	Applied Logic and Digital Design	3 / 6	4 Sequential Circuit - Memory	Flipped Classroom
2	Dr.S.Senthilkumar	29.10.2025	II ECE A	2302EC302 - Digital Electronics	6 / 1	5, Structural Model Programming	Using Modern Tool Usage - Xilinx
3	Dr.RS.Koteeshwari, AP	29.10.2025	III ECE B	2302EC501 – Digital Communication	6 / 4	3 BPSK-BER Performance Analysis	Using Modern Tool Usage - MATLAB
4	Dr.RS.Koteeshwari, AP, Mrs.K.Vembarasi	23.10.2025	III ECE A	2302EC551 - Communication Systems Lab	5 / 2,3,4	Observation – 5 Experiments completed & signed without filling Rubrics	Records – 2 Experiments only signed
5	Dr.M.Irsha Ahamed,	6.11.2025	I IT	2301GEX05 - Applied Digital	6	1 & 2, Tabulation	Debate

	Professor			Logic and Design		Method Vs K-Map	
6	Dr.M.Malathi	7.11.2025	III ECE A	2302EC503 – Transmission Lines and Waveguides	6	4, Impossibility of TEM waves in waveguides	Video Presentation
7	Mrs. R. Keerthika	29.10.2025	III ECE A	2302EC502 - Networks and Security	4	4 Firewall	Modern tool usage (SISCO Packet Tracer)
8	Mrs. R. Mythily	04.11.2025	III ECE (PE)	2303EC001 - Wearable Devices	3	4 Smart Fabrics	Flipped Class Room
9	Mrs. M. Vijaya Durga	04.11.2025	Open Elective	1903EC004 - Display Systems	5	4 Types of Display	Demonstration
10	Mr. Mathiyarasan	04.11.2025	I CSE C	2301GEX05 - Applied Digital Logic and Design	2	2 Encoder Decoder	Quiz



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B.E. – Electronics and Communication Engineering

Third Semester

2302EC302 – Digital Electronics

Modern Tool Usage - Xilinx

29.10.2024

Introduction:

As part of the continuous effort to enhance the teaching-learning process and provide students with practical exposure to modern digital design methodologies, Xilinx software was utilized as an instructional tool for teaching Hardware Description Language (HDL) programming to second-year ECE students. The activity aimed to bridge the gap between theoretical concepts and practical implementation in digital electronics.

Objectives

- ✓ To introduce students to modern digital design and simulation tools.
- ✓ To develop programming skills in HDL such as Verilog/VHDL.

Activity Description

The training sessions were conducted for second-year ECE students as part of the Digital Electronics course. Siliings software was introduced as a platform for writing, simulating, and analyzing HDL programs. The students were guided through the software interface, project creation process, coding practices, simulation techniques, and waveform analysis.

Topics:

- ✓ Introduction to HDL programming concepts.
- ✓ Overview of the Siliings software environment.
- ✓ Development of basic logic gates using HDL.
- ✓ Design and simulation of combinational circuits such as multiplexers, decoders, encoders, and adders.
- ✓ Implementation of sequential circuits including flip-flops, counters, and shift registers.
- ✓ Functional verification using simulation waveforms.
- ✓ Debugging techniques and optimization of HDL code.

Students actively participated in laboratory exercises and completed several design assignments using the software.

Outcomes

- ✓ The activity produced several positive outcomes:
- ✓ Students gained practical knowledge of HDL programming.
- ✓ Enhanced understanding of digital circuit design and verification processes.

Student Feedback

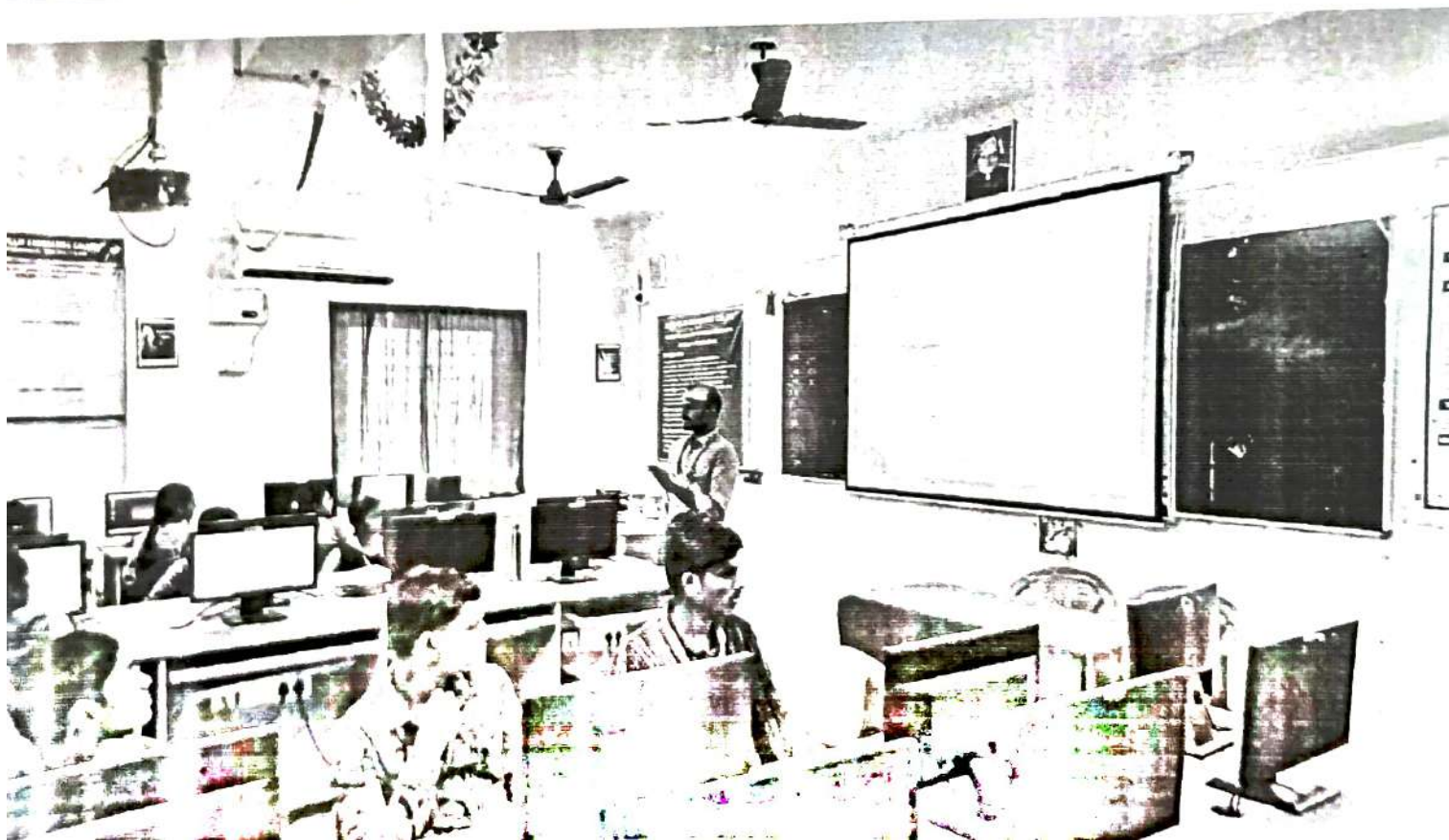
The students expressed positive feedback regarding the use of Xilinx software. They found the software user-friendly and effective in visualizing digital circuit behavior through simulation waveforms. The hands-on approach helped them understand complex digital design concepts more effectively than traditional teaching methods.



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Sample photos





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Academic Year: 2025-26 | Odd Semester | 1st Semester B.E Programs | Regulation: 2023

2301GEX05- Applied Digital logic Design K-Map Vs. Tabulation method

Module 1– 25-10-2025

First-year IT students were assigned the topic "K-Map vs. Tabulation Method" to enhance their understanding of Boolean function minimization techniques. The activity promoted self-learning, analytical thinking, technical presentation skills, and deeper conceptual knowledge in Digital Electronics





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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

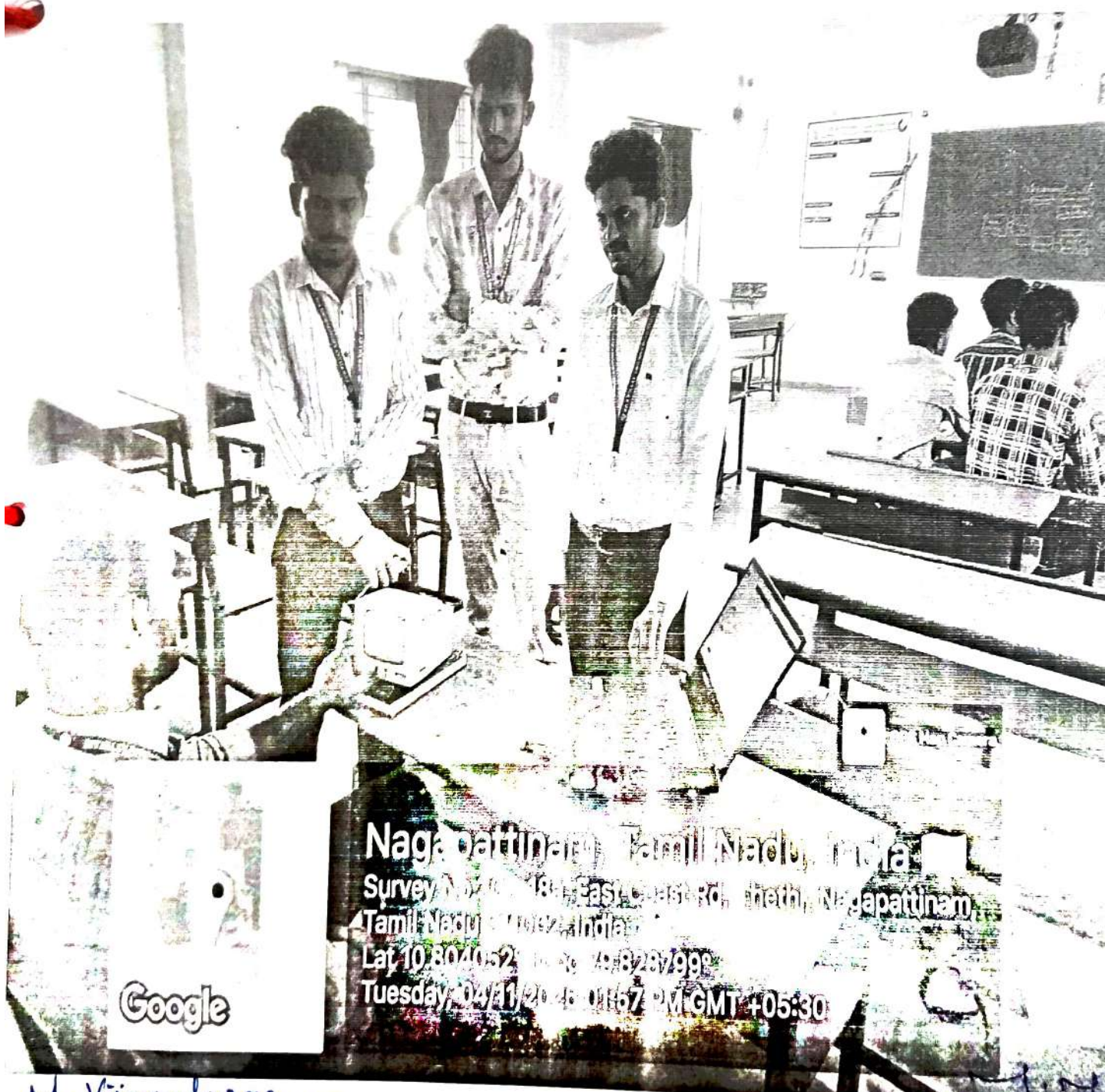
Academic Year: 2025-26 | Odd Semester | 7th Semester B.E Programs | Regulation: 2019

1902EC004 – Display system

Module 4 – Types of display – 04.11.2025

Real time example – Demonstration of different types of display

The demonstration of different types of display systems is an important aspect of **modern pedagogy**, as it educators understand and effectively use technology-based teaching tools. Display systems enhance class communication by presenting information in a clear, attractive, and interactive manner.



M. Vijayadurga
Mrs. M. VIJAYADURGA, M.Tech.,
Assistant Professor
Department of Electronics and Communication Engineering
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Dr. M. MALATHI
Professor & Head / E
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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

Academic Year: 2025-26 | ODD Semester | 1st Semester B.E Programs | Regulation: 2023

2301GEX05 – APPLIED DIGITAL LOGIC & B DESIGN

Module 2 – ENCODER – 22.11.2025

Conducting Quiz – Encoder



M. M. MALATHI
Professor & Head / ECE
E.G.S. Pillay Engineering College
Nagapattinam - 611 002.



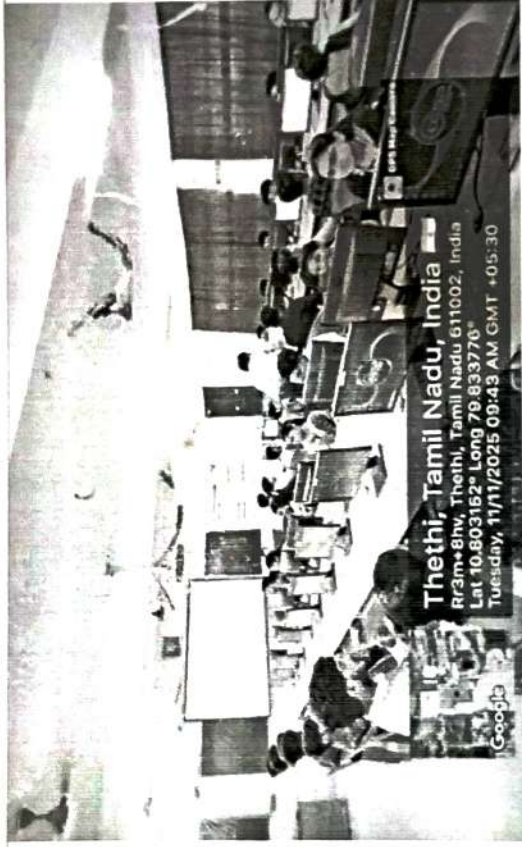
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Interactive Teaching Learning Process

Department of Electronics and Communication Engineering

#	Criteria	Description / Details																										
1.	Title	Innovative Teaching Learning Process																										
2.	Course/Program Specific	Modern Tool Usage																										
3.	Course Code Course Name	2302EC501 –Digital Communication																										
4.	Class/Year/Semester	ECE/III/A/V																										
5.	Name of the Faculty Coordinator	Dr.RS. Koteeshwari																										
6.	Module #: Course Content	BER performance analysis of BPSK,QPSK and DPSK																										
7.	Course Outcome [CO#]	CO 3: Analyze and Design different types of passband modulation techniques																										
8.	CO (Activity) - PO/PSO Mapping	<table><tr><td>PO1</td><td>PO2</td><td>PO3</td><td>PO4</td><td>PO5</td><td>PO6</td><td>PO7</td><td>PO8</td><td>PO9</td><td>PO10</td><td>PO11</td><td>PSO1</td><td>PSO2</td></tr><tr><td>3</td><td>3</td><td>3</td><td>2</td><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>2</td></tr></table>	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	3	3	3	2	2								2
PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2																
3	3	3	2	2								2																
10.	Name of the Expert (if	Nil																										

11.	Position/Company	Nil
12.	Date / Session / Duration / Venue / Mode	11-11-2025/FN/I Hour/SJB 215/Off-Line mode
13.	Objective(s) of the Session	To understand the performance of pass band modulation techniques
14.	Expected Outcome[s]	Able to explain the concepts of basic pass band modulation techniques
15.	Pedagogy	Active Learning
16.	Photos	
17.	Overall Impact (Analysis), upon completion of the Programme	Quantitative outcomes : 61 students have benefited.
18.		Qualitative outcomes : Students can able to explain the performance of basic pass band modulation techniques

Signature
of the Faculty

ESHWARI, M.E., Ph.D.,
Associate Professor
of Electronics and
Communication Engineering
Engineering College (Autonomous).

Signature

Signature of the HOD

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

Academic Year: 2025-26 | Odd Semester | 5th Semester B.E Programs | Regulation: 2023

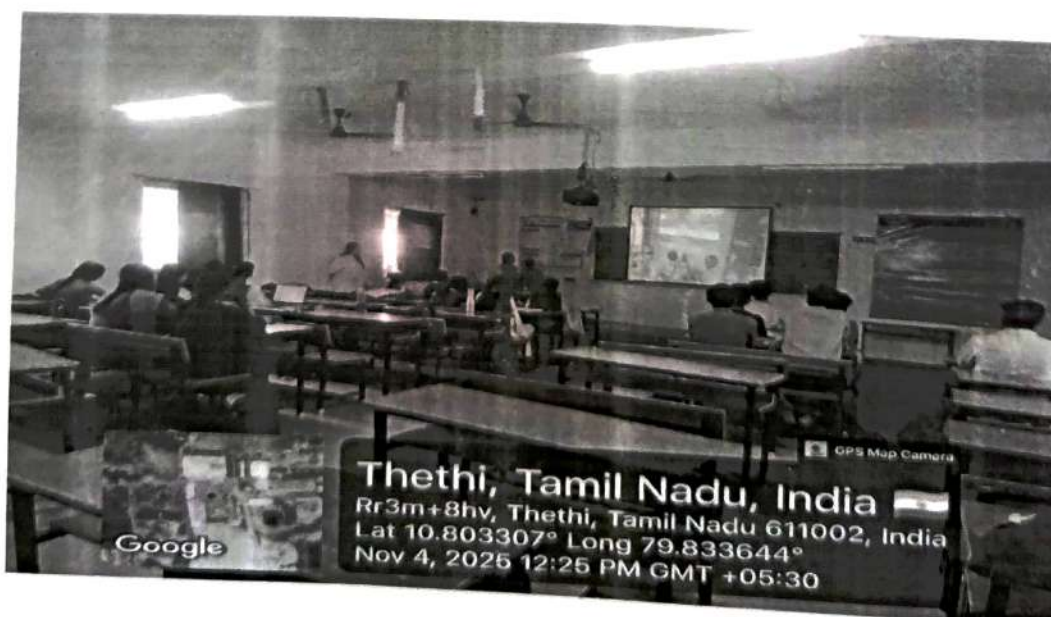
2302EC502 – Wearable Device Flipped Classroom on 04.11.2025 Module 4 – Smart Fabrics

Description:

The flipped classroom is a student-centered teaching method in which students learn basic concepts before class through videos, reading materials, or online resources. Classroom time is then used for discussions, problem-solving, case studies, and hands-on activities.

Implementation in Wearable Devices Course:

- Students are provided with pre-class materials such as videos on wearable sensors, Body Area Networks (BAN), and healthcare applications.
- Students study the materials before attending class.
- During class, students participate in group discussions, quizzes, and case studies on wearable devices.
- Hands-on activities such as analyzing smart watches, fitness bands, or wearable health monitoring systems are conducted.
- The faculty clarifies doubts and guides students in applying concepts to real-world scenarios






Coordinator 5/9/26

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Academic Year: 2025-26 | Odd Semester | 5th Semester B.E Programs | Regulation: 2023

2302EC502 – Networks and Security

Modern Tool Usage - SISCO Packet Tracer on 29.10.2025

Module 4 - Firewall

Computer networks play a vital role in modern communication systems. Security is an essential requirement to protect data and network resources from unauthorized access. This project demonstrates the design and implementation of a secure campus network using Cisco Packet Tracer. The network consists of multiple departments connected through switches and routers with VLAN segmentation, IP addressing, and basic security configurations. The simulation helps understand network management, communication protocols, and security mechanisms in a practical environment.



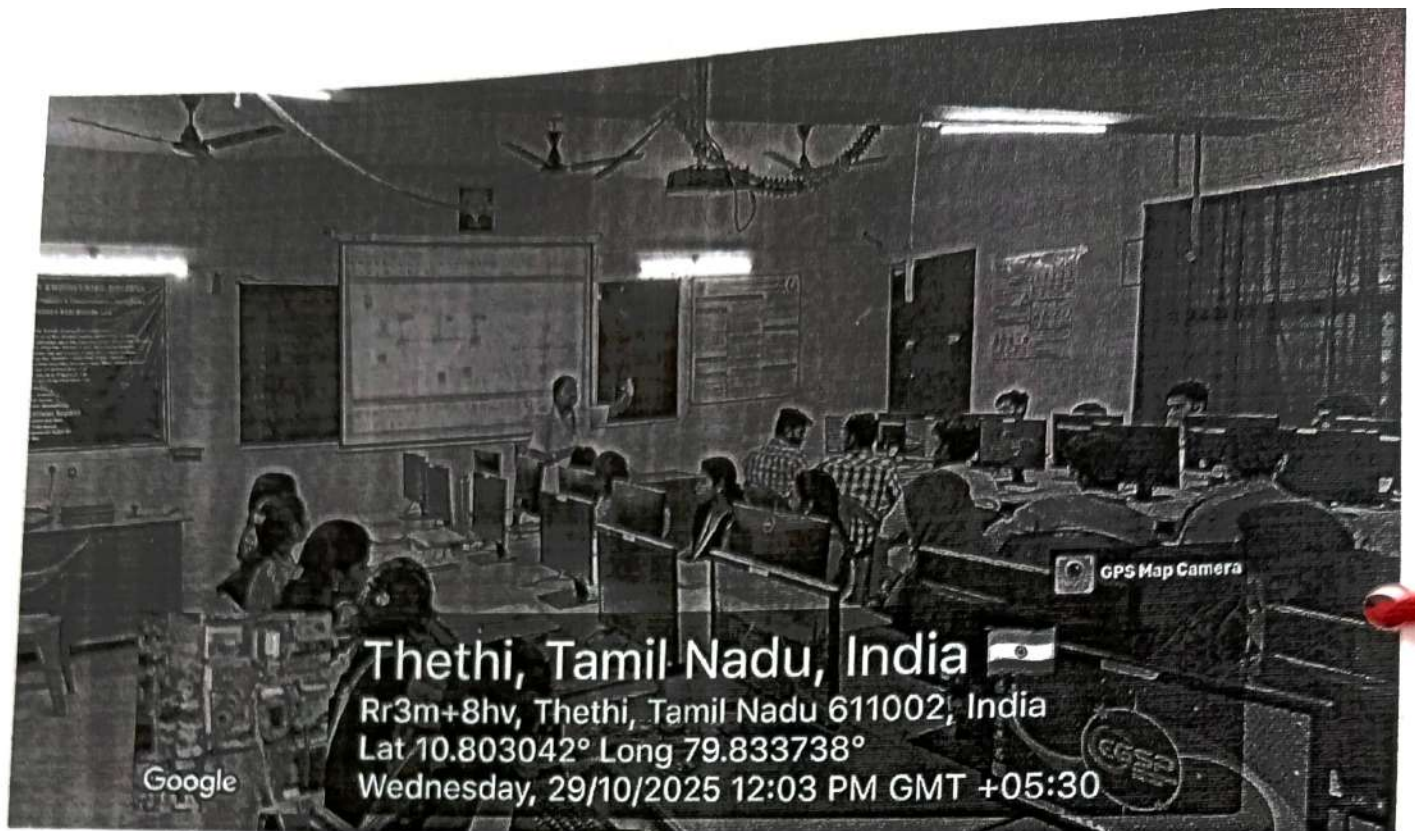
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30/10/25
Course Coordinator

Mrs. R. KEERTHIKA, M.Tech.,
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Nagapattinam - 611 002

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HOD/ECE
Dr. M. MALATHI
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B.E. – Electronics and Communication Engineering

2302EC503 – Transmission Lines and Waveguides

Innovative Practice 1 – ACTIVITY BASED LEARNING

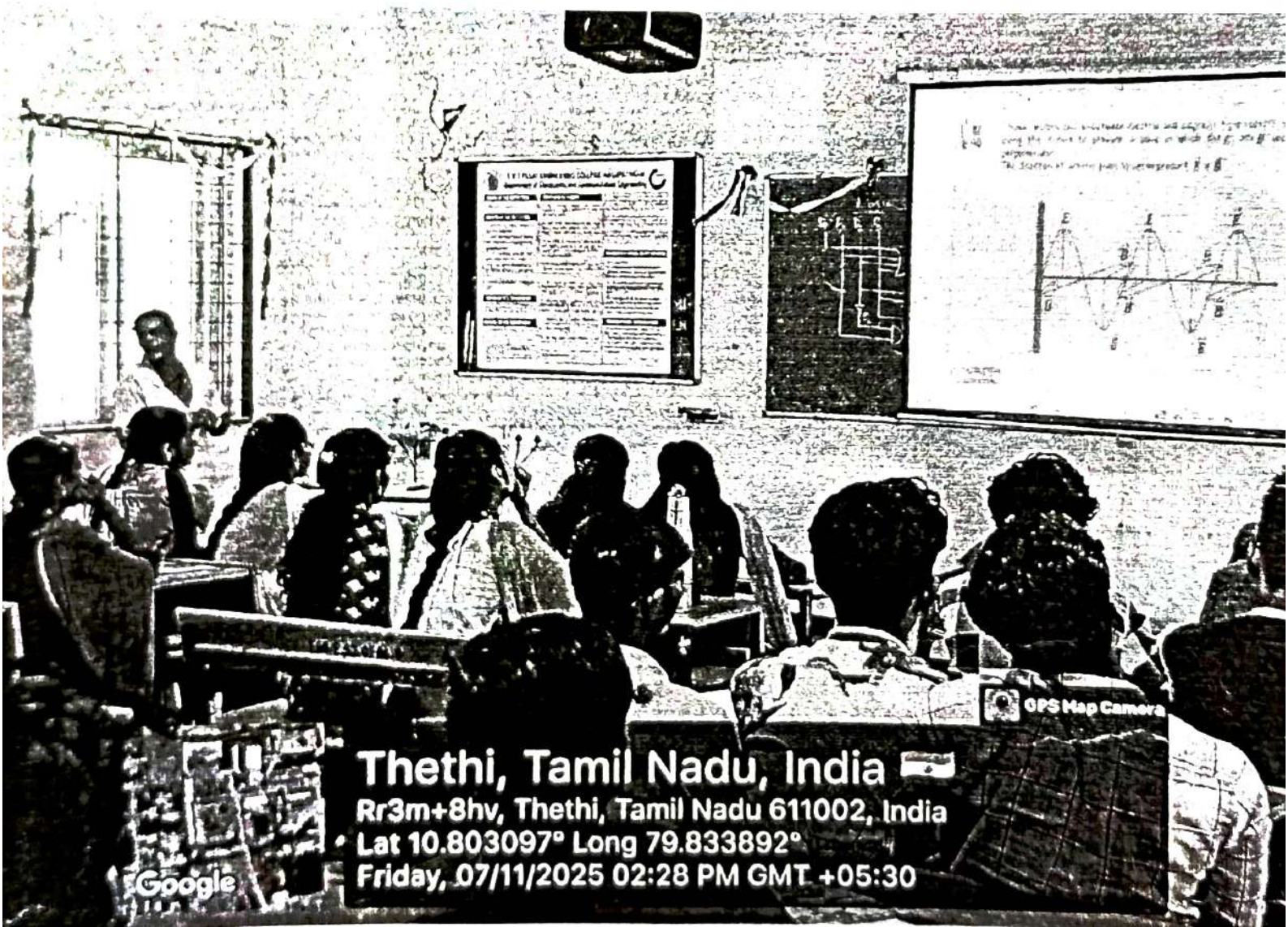
VIDEO PRESENTATION

Innovative video teaching tools transform passive viewing into active learning through **interactive elements**, personalized paths, alongside creation tools for screen/camera recording, and immersive experiences (VR/AR) for complex concepts, all enhancing engagement and understanding beyond traditional lectures

Topics Discussed:

- MODULE 3 – EM wave propagation
- MODULE 4 – Why TEM wave not propagated in Rectangular waveguides

S. No.	Topics	Link	Section	Date
1	EM wave propagation	https://www.google.com/search?q=how+the+EM+waves+propagated+in+waveguides+animation+video+youtube&sca_esv=2321425f395f450e&ei=AfMMaYaCG9mzvr0P5eHL2Ak&ved=0ahUKEwjGhuO3nd6QAxXZma8BHeXwEpsQ4dUDCBE&uact=5&oq=how+the+EM+waves+propagated+in+waveguides+animation+video+youtube&gs_lp=Egxnd3Mtd2l6LXNlcAIQWhvdyB0aGUgRU0gd2F2ZXMcHJvcGFuYXRlZCBpb3YXZlZ3VpZGVzIGFuaW1hdGlvb2aWRIbyB5b3V0dWJlMgQQIRgKSLUjUNkdW0oXcAF4AZABAJgBpQGgAdwHqgEDMC44uAEDyAEA-AEBmAlGgKvBclCChAAGLADGNYEGEfCAGgOABiABBiiBMICBRAAGO8FmAMAiAYBkAYIkgcDMS4loAfdl7IHazAuNbgHqgXCBwewLjluMy4xyAcc&scient=gws-wiz-serp#fpstate=ive&vld=cid:32ff20af,vid:ITjSdnEcJV8,st:0	III ECE A & B	7.11.2025
2	Why TEM wave not propagated in Rectangular waveguides	https://www.youtube.com/watch?v=uDGKJ19gxBM	III ECE A & B	7.11.2025





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B.E. – Electronics and Communication Engineering

2302EC503 – Transmission Lines and Waveguides

Innovative Practice 2 – MODERN TOOL - SOFTWARE

CST STUDIO SUITE

CST Studio Suite is a powerful 3D electromagnetic (EM) simulation software from Dassault Systems SIMULIA, used by engineers and researchers to design, analyze, and optimize EM components like antennas, motors, sensors, and PCBs, by integrating various solver technologies (FEM, FIT, TLM) in a single environment to virtually prototype devices, reduce physical prototypes, and understand complex EM phenomena.

Topics discussed

S. No.	Topics	Module	Section	Date
1	Design mode characteristics	4	III ECE A & B	13.11.2025
2	Design 50 ohm Microstrip Line	5	III ECE A & B	13.11.2025

